

MECHANICAL INTEGRITY - INSPECTION REPORT

Report No: IR-2026-0147 Date of inspection: 12 Aug 2026
Equipment tag: E-101 Description: Shell & tube heat exchanger (crude preheat)
Inspection type: Turnaround internal inspection + UT thickness survey
Design code: TEMA R / ASME VIII Div. 1 Shell material: SA-516 Gr. 70
Nominal shell thickness: 12.0 mm Minimum required thickness (t-min): 8.4 mm
Corrosion allowance: 3.0 mm Service: crude oil, 180 C, 14 barg

1. UT THICKNESS READINGS (SHELL)

CML	LOCATION	READING (mm)
CML-1	Shell top, inlet end	11.4
CML-2	Shell bottom	10.1
CML-3	Near N2 nozzle	9.2
CML-4	Shell mid-span	10.8
CML-5	Outlet end	11.1
CML-6	Bottom, 6 o'clock	9.6

2. VISUAL OBSERVATIONS

- Localised pitting on the shell bottom near the N2 nozzle, maximum pit depth 1.1 mm.
- Tube sheet: minor erosion at tube inlets; 14 tubes previously plugged.
- Gasket seating surface at the channel flange is satisfactory.
- External coating intact; no corrosion under insulation observed at inspection ports.

3. HYDROSTATIC / LEAK TEST

Tube side test pressure 21.0 barg held 30 min - no leak observed.

Shell side test pressure 18.2 barg held 30 min - no leak observed.

4. ASSESSMENT

Minimum measured thickness is 9.2 mm at CML-3, against t-min of 8.4 mm.

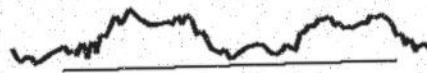
Remaining corrosion margin is 0.8 mm. Measured corrosion rate is 0.25 mm/year (comparison with the 2022 survey). Estimated remaining life at CML-3: 3.2 years.

5. RECOMMENDATION

Fit for continued service until the next turnaround (2028), subject to:

- (a) re-inspection of CML-3 within 12 months;
- (b) replacement of 6 further tubes showing wall loss greater than 40%.

Inspected by: A. Deshmukh (Level II UT)



Reviewed by: QA - Inspection Cell

